

Product Sales Data Analysis Report

A Python (Pandas / Matplotlib / Seaborn) exploratory data analysis project

1. Project Overview

This project analyzes product sales records containing daily records for four products: P1, P2, P3 and P4. The goal is to unfold the insights of the product sales dataset according to sales, quantity, trends and numerical relationships.

2. Dataset Summary

The dataset consists of:

- Rows: 4,600
- Columns: 10 original columns
- Date range: 13-06-2010 to 03-02-2023
- Products: P1, P2, P3 and P4
- Sales columns: S-P1, S-P2, S-P3, S-P4
- Quantity columns: Q-P1, Q-P2, Q-P3, Q-P4
- Unnamed: 0 is an additional index column
- There are 0 missing values in the dataset

3. Exploratory Data Analysis (EDA)

The EDA is done using Python and its libraries. Pandas is used for data cleaning, and Matplotlib and Seaborn are used for data visualization.

1. Imported the dataset and important libraries

```
# import the important packages
import pandas as pd # library used for data manipulation and analysis
import numpy as np  # library used for working with arrays
import matplotlib.pyplot as plt # library for plots and visualizations
import seaborn as sns # library for visualization
# To ignore warnings
import warnings
warnings.filterwarnings("ignore")
```

2. Loaded the dataset and checked the first/last records, information and shape

```
data= pd.read_csv('statsfinal.csv')
```

```
data.head()
data.tail()
data.info()
data.shape
```

3. Checked the dataset description and summary statistics

```
data.describe()  
data.info()  
df.head()  
df.tail()
```

4. Checked the null values, duplicates and unique values

```
#missing value check  
df.isnull().sum()
```

```
#duplicated row check  
df.duplicated().sum()
```

```
df.nunique()
```

```
df['Q-P1'].unique()
```

```
df['Q-P4'].unique()
```

```
df['Q-P1'].value_counts()
```

```
df.isna().sum()
```

There are no missing value

5. Checked data types and converted Date to category

```
df.dtypes  
df.info()  
df['Date'] = df['Date'].astype('category')  
df.info()  
df.describe()  
df.columns  
df.isnull()
```

6. Removed the unwanted index column and checked duplicates

```
#remove unwanted column
df.drop(columns=['Unnamed: 0'], errors = 'ignore', inplace = True)
```

```
df.columns
```

```
#check duplicate value
df.duplicated().sum()
```

7. Calculated total sales and total quantity

```
sales_columns = [
    'S-P1',
    'S-P2',
    'S-P3',
    'S-P4'
]

quantity_columns = [
    'Q-P1',
    'Q-P2',
    'Q-P3',
    'Q-P4'
]

df['Total_Sales'] = df[sales_columns].sum(axis=1)

df['Total_Quantity'] = df[quantity_columns].sum(axis=1)
```

8. Created the product performance summary

```
product = pd.DataFrame({

    'Product':
    ['P1', 'P2', 'P3', 'P4'],

    'Total Sales':
    [
        df['S-P1'].sum(),
        df['S-P2'].sum(),
        df['S-P3'].sum(),
        df['S-P4'].sum()
    ],

    'Total Quantity':
    [
        df['Q-P1'].sum(),
        df['Q-P2'].sum(),
        df['Q-P3'].sum(),
        df['Q-P4'].sum()
    ]

})
product
```

4. Data Visualization

The data visualization is done using Matplotlib and Seaborn in Jupyter after data cleaning with Pandas.

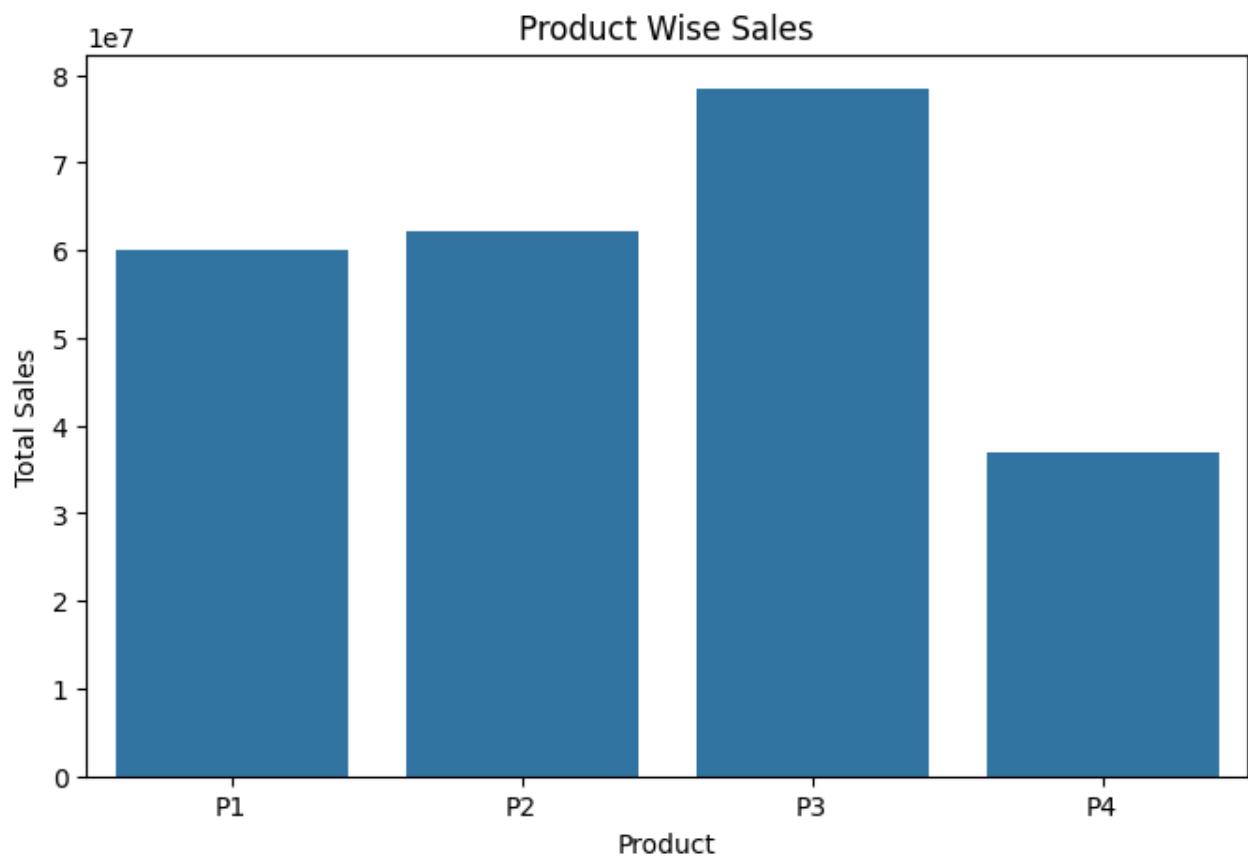
1. Product-wise sales distribution

```
plt.figure(figsize=(8,5))

sns.barplot(
    data=product,
    x='Product',
    y='Total Sales'
)

plt.title("Product Wise Sales")

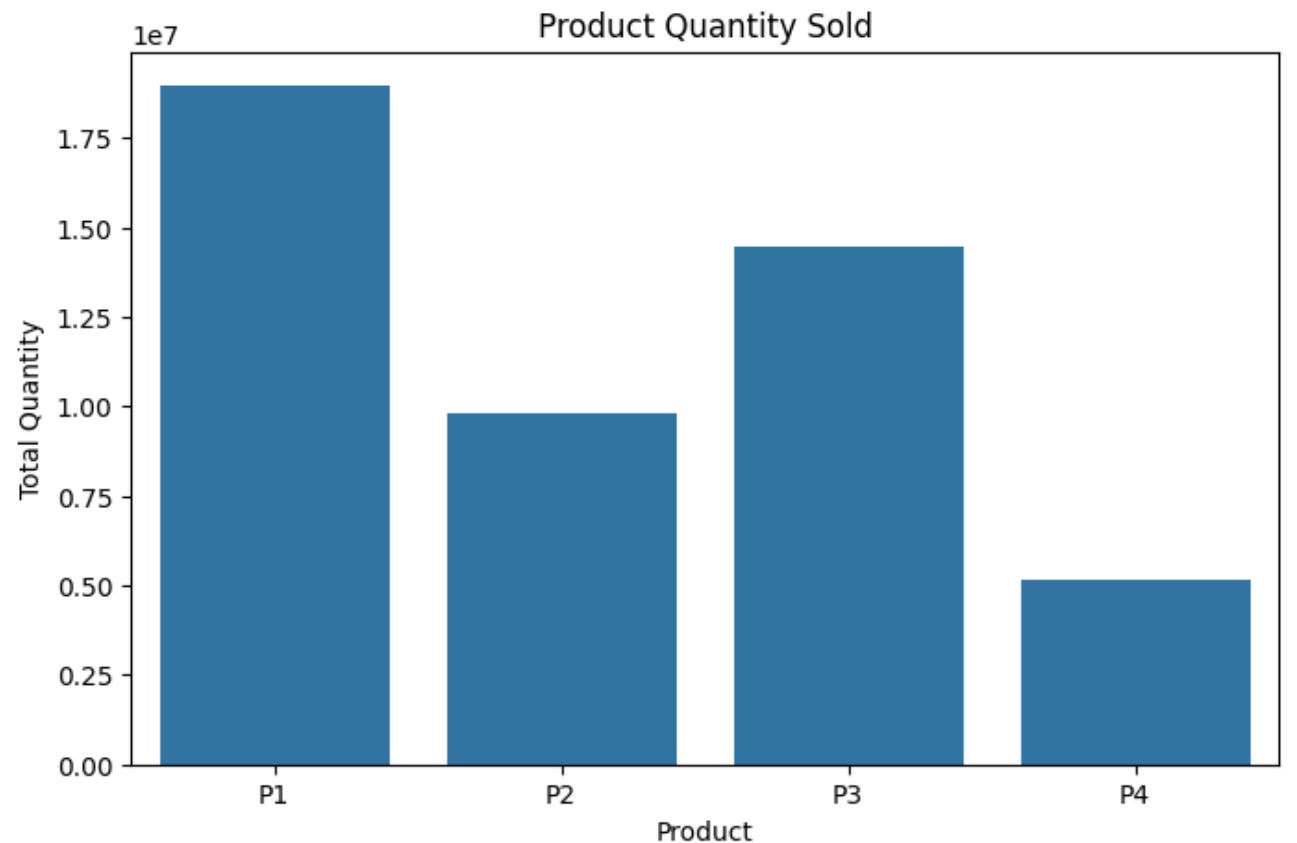
plt.show()
```



Interpretation: P3 has the highest total sales, followed by P2 and P1, while P4 has the lowest total sales.

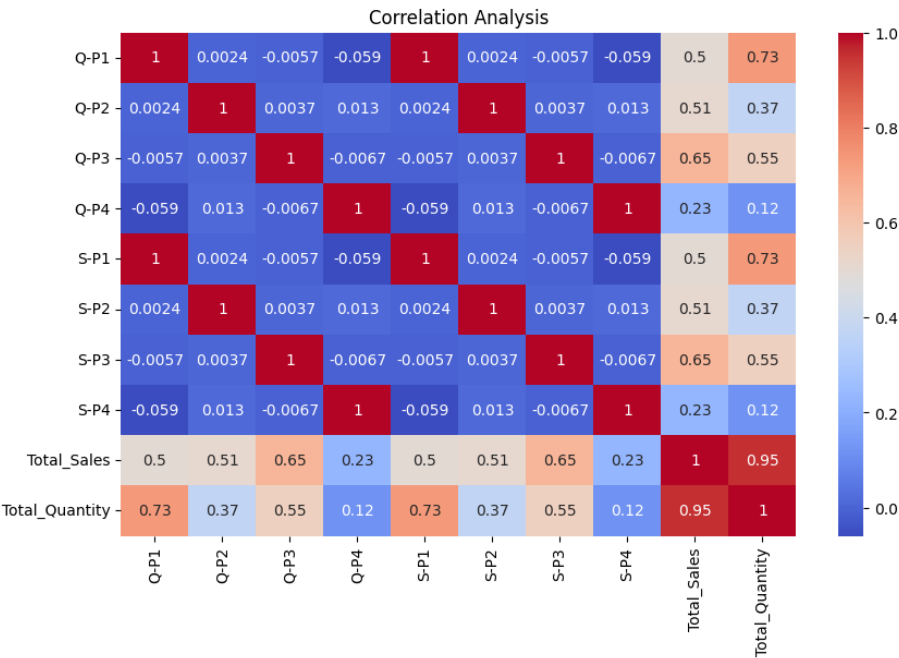
2. Product quantity sold

```
plt.figure(figsize=(8,5))
sns.barplot(
    data=product,
    x='Product',
    y='Total Quantity'
)
plt.title(
    "Product Quantity Sold"
)
plt.show()
```



Interpretation: P1 has the highest total quantity sold, while P4 has the lowest quantity sold.

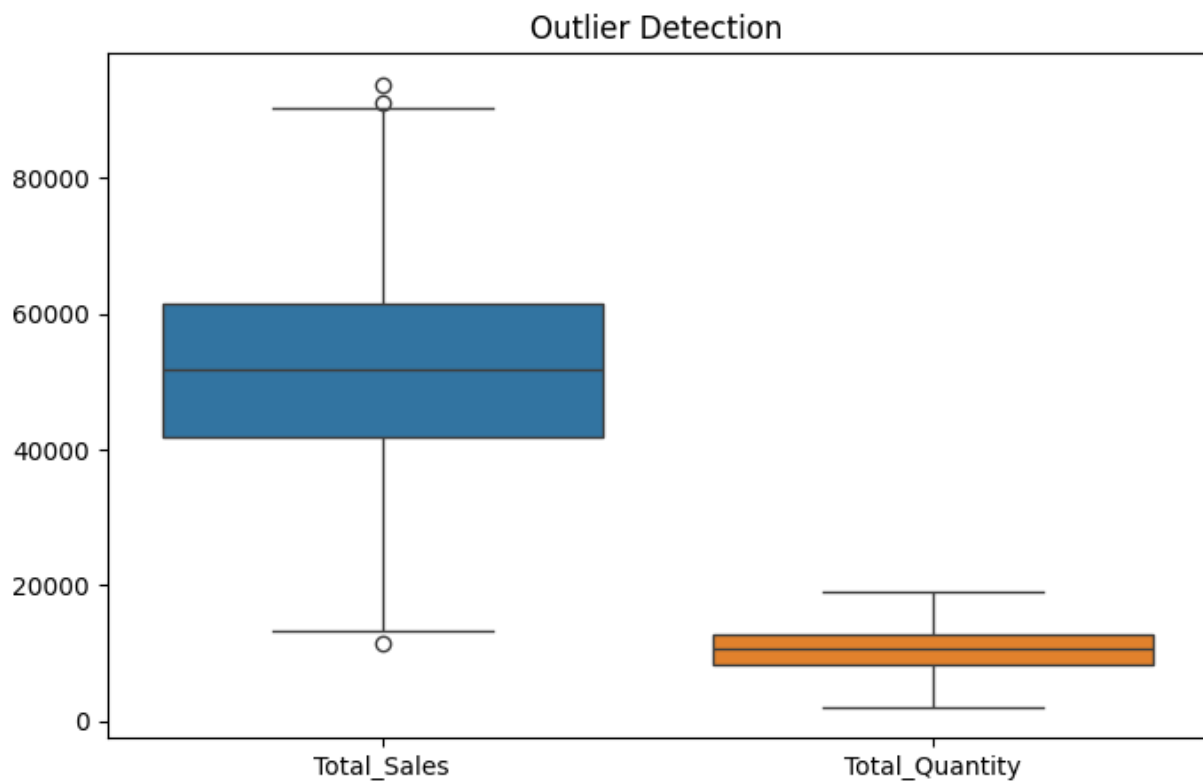
3. Correlation analysis



4. Outlier detection

```
plt.figure(figsize=(8,5))
sns.boxplot(
    data=df[['Total_Sales', 'Total_Quantity']])
plt.title("Outlier Detection")

plt.show()
```



Interpretation: The box plot is used to inspect the distribution and potential outliers of Total_Sales and Total_Quantity.

5. Monthly sales trend code from the notebook

```
df['Date'] = pd.to_datetime(df['Date'], errors='ignore')
```

```
monthly_sales = (
    df.groupby(
        df['Date'].dt.to_period('M')
    )
    ['Total_Sales']
    .sum()
)
plt.figure(figsize=(12,5))
monthly_sales.plot(
    marker='o'
)

plt.title(
    "Monthly Sales Trend"
)

plt.xlabel("Month")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.show()
```

Note: The source notebook raises an `AttributeError` at this step because `Date` is not `datetimelike` after the earlier category conversion. Therefore, no monthly-sales graph is inserted.

Final business insight code

```
best_product = product.loc[
    product['Total Sales'].idxmax()
]

final_report = pd.DataFrame({
    'Metric':
    [
        'Total Revenue',
        'Total Quantity Sold',
        'Average Sales',
        'Best Selling Product'
    ],
    'Value':
    [
        df['Total_Sales'].sum(),
        df['Total_Quantity'].sum(),
        df['Total_Sales'].mean(),
        best_product['Product']
    ]
})
final_report
```

Insight from the final business insight code

The final business insight calculation summarizes the overall performance of the dataset.

Total Revenue: 237,510,477.00

Total Quantity Sold: 48,398,305

Average Sales: 51,632.71

Best Selling Product: P3

This shows that P3 has the highest total sales among the four products, while the overall dataset records 48,398,305 units sold. The average total sales per record is 51,632.71.

5. Conclusion

Q1: Which product has the highest total sales?

P3 has the highest total sales, with total sales of 78,429,589.68.

Q2: Which product has the highest total quantity sold?

P1 has the highest total quantity sold, with a total quantity of 18,960,506.

Q3: Which product has the lowest sales and quantity?

P4 has the lowest total sales at 36,848,553.00 and the lowest total quantity at 5,168,100.

Q4: What is the total revenue and total quantity sold?

The total revenue is 237,510,477.00 and the total quantity sold is 48,398,305.

Q5: What is the best-selling product according to total sales?

According to the final business insight calculation, P3 is the best-selling product by total sales.

Summary

The product sales analysis shows clear differences between sales value and quantity sold. P3 generates the highest total sales, while P1 records the highest total quantity. P2 ranks next in total sales, and P4 records the lowest sales and quantity. The analysis also includes data-quality checks, product-level calculations, correlation analysis and outlier detection. The monthly sales trend code is retained exactly from the source notebook, but its graph is not included because the original notebook cell fails after the Date column has been converted to category.